4.9 Spectroscopy and chromatography

Knowledge of the concepts introduced in *Unit 2, Topic 2.12: Mass Spectra and IR* will be assumed and extended in this topic.

Students will be assessed on their ability to:

a explain the effect of different types of radiation on molecules and how the principles of this are used in chemical analysis and in reactions, limited to:

  i infrared in analysis

  ii microwaves for heating

  iii radio waves in nmr

  iv ultraviolet in initiation of reactions

b explain the use of high resolution nmr spectra to identify the structure of a molecule:

  i based on the different types of proton present from chemical shift values

  ii by using the spin-spin coupling pattern to identify the number of protons adjacent to a given proton

  iii the effect of radio waves on proton spin in nmr, limited to ^{1}H nuclei

  iv the use of magnetic resonance imaging as a non-invasive technique, eg scanning for brain disorders, or the use of nmr to check the purity of a compound in the pharmaceutical industry

c demonstrate an understanding of the use of IR spectra to follow the progress of a reaction involving change of functional groups, eg in the chemical industry to determine the extent of the reaction

d interpret simple mass spectra to suggest possible structures of a simple compound from the *m/e* of the molecular ion and fragmentation patterns

e describe the principles of gas chromatography and HPLC as used as methods of separation of mixtures, prior to further analysis (theory of R_f values not required), and also to determine if substances are present in industrial chemical processes.